

AP Statistics MCQ Practice Exam 11

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PART A, QUESTIONS

APSTAT-MCQ11-001

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

In an ordered dataset of 20 observations, positions 12 through 15 all contain the value 7, position 11 is less than 7, and position 16 is greater than 7. Which pair of percentages is correct?

- A) 60% are at most 7, and 55% are below 7.
- B) 75% are below 7, and 55% are at most 7.
- C) 55% are at most 7, and 75% are below 7.
- D) 75% are at most 7, and 55% are below 7.

APSTAT-MCQ11-002

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A 95% confidence interval for $p_A - p_B$, the difference in signup proportions for two workflows, is (0.02, 0.10). Before the study, a difference of at least 0.08 was defined as practically important. Which conclusion is justified?

- A) There is no evidence of any positive difference because 0.08 lies in the interval.
- B) The population difference is exactly 0.06, the midpoint of the interval.
- C) The data support a positive difference, but the interval does not establish that the difference is at least 0.08.
- D) Every plausible difference is practically important because the interval excludes 0.

APSTAT-MCQ11-003

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Three volunteers randomly receive three distinct task cards labeled with their own names, with all 3! assignments equally likely. What is the probability that no volunteer receives the card bearing that volunteer's name?

- A) $1/6$
- B) $1/3$
- C) $1/2$
- D) $2/3$

APSTAT-MCQ11-004

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

Researchers compare mean recovery time after two warm-up routines. Baseline running speed is expected to affect recovery, and each runner can use only one routine. Which design best controls baseline speed?

- A) Form pairs of runners with similar baseline speeds, then randomly assign one runner in each pair to each routine and analyze within-pair recovery-time differences.
- B) Assign all faster runners to one routine and all slower runners to the other.
- C) Let runners choose the routine they expect will work best.
- D) Give every runner the first routine and compare the results with historical records for the second.

APSTAT-MCQ11-005

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

A botanist wants a regression model that uses leaf area to predict transpiration rate for individual plants. Which data record is the appropriate observational unit?

- A) One list of leaf areas and a separate list of transpiration rates sorted independently
- B) One average leaf area and one average transpiration rate for the entire greenhouse
- C) A category indicating whether each greenhouse contains large or small plants

D) A paired leaf-area and transpiration-rate measurement from each sampled plant

APSTAT-MCQ11-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

An organizer wants to estimate satisfaction among all holders of tickets to a three-day festival, including no-shows and people who leave early. Which plan provides the best sampling frame?

- A) Survey people still present during the final performance.
- B) Post an open survey on the performer's fan page.
- C) Draw an SRS from the full ticket-holder database and contact the selected people after the festival.
- D) Interview the first 100 people who use the main exit.

APSTAT-MCQ11-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A sample of 250 transactions has $\hat{p} = 0.28$, and a reported proportion interval has margin of error 0.056. Which pair correctly gives the number of sampled successes and the interval?

- A) 28 successes and (0.224, 0.336)
- B) 70 successes and (0.168, 0.392)
- C) 72 successes and (0.224, 0.336)
- D) 70 successes and (0.224, 0.336)

APSTAT-MCQ11-008

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

The cumulative distribution function of a discrete random variable X has $F(2) = 0.45$, while the probability accumulated strictly below 2 is $P(X < 2) = 0.30$. What does the jump at $x = 2$ represent?

- A) $P(X > 2) = 0.15$
- B) $P(X \leq 2) = 0.15$

C) $P(X = 2) = 0.15$

D) $P(X < 2) = 0.45$

APSTAT-MCQ11-009

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For two independent sample means, $SE(\bar{x}_1 - \bar{x}_2) = \sqrt{(s_1^2/n_1 + s_2^2/n_2)} = 2.5$ units. If $s_1/\sqrt{n_1} = 2.0$ units, what is $s_2/\sqrt{n_2}$?

A) 0.5 unit

B) 1.5 units

C) 2.25 units

D) 4.5 units

APSTAT-MCQ11-010

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

An empirical cumulative distribution shows that 35% of observations are at or below 10 and 82% are at or below 20. What proportion of observations are greater than 10 and at or below 20?

A) 0.18

B) 0.35

C) 0.82

D) 0.47

APSTAT-MCQ11-011

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

A least-squares regression fitted to a dataset has $SSE = 100$. One observation has residual 6. If that observation is removed and the line is refitted to the remaining data, what can be concluded about the new SSE?

- A) It is at most 64.
- B) It must equal 64.
- C) It must remain 100 because the original line was least squares.
- D) It is at least 136 because the removed residual is squared.

APSTAT-MCQ11-012

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A warranty office wants to estimate the proportion of all claims filed within 30 days of purchase. Which parameter matches that goal?

- A) x , the number of sampled claims filed within 30 days
- B) p , the population proportion of all warranty claims filed within 30 days
- C) $\hat{p}$, the sample proportion filed within 30 days
- D) μ , the population mean number of claims filed per day

APSTAT-MCQ11-013

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

A school wants to estimate the probability that at least two students in a randomly formed group of 25 share a birthday, ignoring leap day and assuming equally likely independent birthdays. Which simulation is appropriate?

- A) Generate 25 different integers from 1 to 365 without replacement and count repeated values.
- B) Generate one integer from 1 to 365 and multiply it by 25.
- C) Generate 365 integers from 1 to 25 and count groups with no repetitions.
- D) Generate 25 integers from 1 to 365 with replacement, record whether any value repeats, and repeat for many trials.

APSTAT-MCQ11-014

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

A dotplot has two tight clusters, one near 10 and the other near 30, with a gap around the mean of 20. Which interpretation is most appropriate?

- A) The mean is an arithmetic balance point but may poorly represent a typical observation when it lies in the gap between clusters.
- B) The mean must be one of the observations, so the described dotplot is impossible.
- C) The gap proves that the standard deviation is 0.
- D) The distribution is unimodal because it has only one mean.

APSTAT-MCQ11-015

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A 95% confidence interval for $\mu_1 - \mu_2$ is (1.2, 5.8), centered at 3.5, and was constructed using $t^* = 2.00$. What is the t statistic for testing $H_0: \mu_1 - \mu_2 = 0$ using the same standard error?

- A) Approximately 3.04
- B) Approximately 1.52
- C) Exactly 2.00
- D) Approximately 0.76

APSTAT-MCQ11-016

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A survey deliberately samples rural households at four times their population rate, then reports the unweighted sample proportion and a simple-SRS confidence interval. Which assessment is correct?

- A) The analysis can misrepresent the population proportion and its uncertainty unless sampling weights and the design are incorporated.
- B) Oversampling automatically makes the unweighted sample proportion unbiased.
- C) The simple-SRS interval is valid because every sampled household answered the same question.
- D) Weighting is unnecessary whenever every subgroup has at least 10 successes and 10 failures.

APSTAT-MCQ11-017

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

A retailer wants to study the distribution of checkout delays, not total store traffic. Which variable and observational unit best match that goal?

- A) Store as the unit and total daily sales as the variable
- B) Cashier as the unit and cashier identification number as the variable
- C) Randomly selected transaction as the unit and seconds from joining the line to beginning checkout as the variable
- D) Day as the unit and whether the store opened on time as the variable

Questions APSTAT-MCQ11-018 to APSTAT-MCQ11-020 refer to the following stimulus.

A calibration procedure passes any one check with probability 0.80. Six checks are performed independently under identical conditions. Let X be the number of checks that pass, so $X \sim \text{Binomial}(6, 0.80)$.

APSTAT-MCQ11-018

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For $X \sim \text{Binomial}(6, 0.80)$, what is $P(X = 6)$?

- A) 0.8
- B) $6(0.8)(0.2)^5$
- C) $0.8^6 = 0.262144$
- D) $1 - 0.8^6$

APSTAT-MCQ11-019

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For $X \sim \text{Binomial}(6, 0.80)$, what is $P(X \geq 5)$?

- A) 0.262144
- B) 0.393216

C) 0.737856

D) $6(0.8^5)(0.2) + 0.8^6 = 0.65536$

APSTAT-MCQ11-020

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Suppose the six-check procedure is repeated independently 1,000 times under the calibration model. Which result is a reasonable expectation based on $P(X \geq 5) = 0.65536$?

- A) Exactly 656 of the first 1,000 individual checks will pass.
- B) Roughly 655 of the 1,000 procedures will record five or six passing checks.
- C) Roughly 800 of the 1,000 procedures will record exactly five passing checks.
- D) All 1,000 procedures will record at least five passing checks.

APSTAT-MCQ11-021

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

For a fitted least-squares line with an intercept, the residuals sum to 0. Which additional conclusion is justified?

- A) Exactly half of the residuals are positive and half are negative.
- B) Every positive residual has a negative residual of the same magnitude.
- C) The median residual must be 0.
- D) Positive and negative residuals balance algebraically, but their counts and individual magnitudes need not match.

APSTAT-MCQ11-022

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A website compares purchase proportions for two banner designs. Some visitors make many visits during the study. Which randomization and recording plan best avoids treating repeat visits as

independent people?

- A) Randomly assign each eligible user to one banner for the study and record one prespecified purchase outcome per user.
- B) Randomly assign a new banner on every page view and treat every view as an independent observation.
- C) Show one banner to weekday visits and the other to weekend visits.
- D) Let frequent visitors choose which banner they see.

APSTAT-MCQ11-023

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

For the ordered data 1, 4, 5, 6, 9, 40, define the trimmed mean by removing the smallest and largest observation and averaging the four remaining values. What is the trimmed mean?

- A) 6
- B) 6.5
- C) 10
- D) 10.83

APSTAT-MCQ11-024

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

Twenty classrooms are randomly assigned to one of two curricula, and the response used for each classroom is its mean end-of-term score. Which parameter most directly matches a comparison that respects the unit of assignment?

- A) The difference between two individual students' scores
- B) The proportion of all students scoring above the combined median
- C) The mean of all 20 classroom means without regard to curriculum
- D) $\mu_A - \mu_B$, the difference in population mean classroom-average scores under curricula A and B

APSTAT-MCQ11-025

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

After standardization, a measurement has $Z \sim N(0, 1)$. Restrict attention to observations with $Z > 0$. What proportion of that restricted group has $Z > 1$? Use $\Phi(1) = 0.8413$.

- A) 0.1587
- B) 0.3174
- C) 0.5000
- D) 0.8413

APSTAT-MCQ11-026

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

An estimated difference in population proportions is 0.080 with standard error 0.030. Candidate two-sided critical values are 1.645 for 90%, 1.960 for 95%, 2.576 for 99%, and 2.807 for 99.5%. What is the lowest listed confidence level whose interval includes 0?

- A) 99%
- B) 99.5%
- C) 95%
- D) 90%

APSTAT-MCQ11-027

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A multilingual health survey needs versions of a questionnaire in several languages. Which procedure best reduces wording-related measurement differences?

- A) Use professional translation and independent back-translation, pilot the versions, and train interviewers to follow equivalent scripts.
- B) Let each interviewer translate questions spontaneously during the interview.
- C) Use different concepts in each language when a literal translation is difficult.
- D) Ask only respondents who are fluent in the original language.

APSTAT-MCQ11-028

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

In an independent-groups analysis of mean curing times, the sample contrast is 8 minutes and the hypothesized contrast is 2 minutes. The two terms contributing to the squared standard error are 4 and 4. What is the resulting t statistic?

- A) 1.41
- B) 2.12
- C) 2.83
- D) 4.24

APSTAT-MCQ11-029

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A 95% confidence interval for $\mu_{\text{new}} - \mu_{\text{standard}}$ in tensile strength is (1.2, 4.8) megapascals. Which conclusion is supported by the interval?

- A) The data support a higher population mean for the new material, although individual strengths from the two materials may overlap.
- B) Every specimen made from the new material is stronger than every standard specimen.
- C) At least 95% of new-material specimens exceed the strongest standard specimen.
- D) The performance gain is necessarily large enough for every engineering application.

APSTAT-MCQ11-030

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A dataset has five-number summary minimum 2, $Q1 = 5$, median 8, $Q3 = 12$, and maximum 30. The maximum is replaced by 12 while all other observations remain unchanged. Which pair of measures is guaranteed for the modified data?

- A) Range = 28 and IQR = 7

- B) Range = 10 and IQR = 10
- C) Range = 10 and IQR = 7
- D) Range = 7 and IQR = 28

APSTAT-MCQ11-031

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

To represent whether a randomly selected account is overdue, which random variable makes the population proportion overdue equal to $E(X)$?

- A) X is the account balance in dollars.
- B) X is the total number of overdue accounts in the population.
- C) $X = 1$ if the selected account is overdue and $X = 0$ otherwise.
- D) X is the number of days until an account first becomes overdue.

APSTAT-MCQ11-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

For an estimate above p_0 , its standardized statistic is $z = 1.74$. A directional rule uses cutoff 1.645, whereas an ordinary 95% interval uses cutoff 1.960. Which statement explains the resulting decisions?

- A) Both procedures must exclude p_0 because 1.74 is positive.
- B) The directional 5% rule rejects p_0 , while the 95% two-sided interval contains p_0 because it uses the larger cutoff.
- C) Neither procedure can use the same estimate and standard error.
- D) The directional rejection assigns probability less than 0.05 to p_0 .

APSTAT-MCQ11-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A unimodal quantitative distribution has mean 50 and median 50. What must be true?

- A) The distribution is exactly symmetric.
- B) The standard deviation is 0.
- C) Half of the observations equal 50.
- D) The two centers are equal, but the distribution can still be asymmetric or otherwise irregular.

APSTAT-MCQ11-034

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

In a paired study with $n = 15$, the before measurements and after measurements are each strongly right-skewed, but the 15 within-pair differences are roughly symmetric with no outliers. Which assessment is most appropriate?

- A) A paired t procedure is invalid because both marginal distributions must be normal.
- B) A paired t procedure can be reasonable because its condition concerns the distribution of within-pair differences.
- C) The data must be analyzed as two independent samples.
- D) A one-proportion z procedure should replace the paired t procedure.

Questions APSTAT-MCQ11-035 to APSTAT-MCQ11-037 refer to the following stimulus.

Twelve distinct paired observations produce the least-squares line $\hat{y} = 4 + 2x$, correlation $r = 0.60$, and $SSE = 96$. A second file is created by copying every row exactly once, so it contains 24 rows but no new measurements.

APSTAT-MCQ11-035

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

The 12 regression pairs give $\hat{y} = 4 + 2x$. After every pair is copied once, what least-squares line is fitted to the 24-row file?

- A) $\hat{y} = 8 + 4x$
- B) $\hat{y} = 4 + x$
- C) $\hat{y} = 2 + 2x$
- D) $\hat{y} = 4 + 2x$

APSTAT-MCQ11-036

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

The 12 original regression pairs have correlation $r = 0.60$. What is the correlation after every pair is copied once?

- A) 0.60
- B) 0.30
- C) 1.20
- D) 0

APSTAT-MCQ11-037

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

The 12 original regression pairs have $SSE = 96$. Every row is then copied once. Which statement about the 24-row file is correct?

- A) SSE remains 96 because the fitted line is unchanged.
- B) SSE becomes 48 because twice as many rows improve the fit.
- C) SSE becomes 192 because every squared residual is counted twice, but the copied rows provide no new independent measurements.
- D) SSE becomes 0 because each copied observation has an identical partner.

APSTAT-MCQ11-038

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For a randomly selected catered order, a shop's profit adjustment Y is +\$6 with probability 0.25, +\$1 with probability 0.50, and -\$4 with probability 0.25. What is $E(Y)$?

- A) \$2.75
- B) \$1.50
- C) \$1.00
- D) -\$1.00

APSTAT-MCQ11-039

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A two-sided z-test of a population proportion has p-value 0.04. Ignoring rounding, at what two-sided confidence level would the null value lie exactly at an endpoint of the corresponding confidence interval?

- A) 96%
- B) 95%
- C) 98%
- D) 99%

APSTAT-MCQ11-040

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

Two research teams estimate comparable effects with nearly identical precision. Their reported two-sided tail areas under their null models are 0.048 and 0.052. What is the soundest synthesis?

- A) Only the first team has demonstrated that its claimed effect is real.
- B) Crossing 0.05 makes the second result substantially weaker than the first.
- C) The data provide nearly the same strength of evidence; a 0.05 convention changes the formal labels, not the evidence abruptly.
- D) The second result supplies more than twice as much support for its null model.

APSTAT-MCQ11-041

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A factory wants an overall estimate of a package characteristic, and output volume differs greatly across three work shifts. Which plan best represents production?

- A) Inspect the same number of conveniently chosen packages from every shift and ignore shift output.

- B) Stratify by shift and randomly sample packages in proportion to each shift's share of total output.
- C) Inspect only the largest shift because it produces most packages.
- D) Ask shift supervisors to submit packages they consider typical.

APSTAT-MCQ11-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A study tests whether a scent affects concentration scores. Which design best controls expectation effects?

- A) Let participants choose whether to use the scented diffuser.
- B) Randomly assign participants to indistinguishable diffusers containing the scent or an odor-matched inactive control, and blind participants and outcome scorers when feasible.
- C) Give the scented diffuser to morning participants and no diffuser to afternoon participants.
- D) Tell all participants that the diffuser is expected to improve concentration and compare scores with last year's group.

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ11-001, Correct: D, Correct Answer: 75% are at most 7, and 55% are below 7., Explanation: Position 15 is 7, so $15/20 = 75\%$ of observations are at most 7. Position 11 is the last value below 7, so $11/20 = 55\%$ are strictly below 7; ties make the two percentages different.

APSTAT-MCQ11-002, Correct: C, Correct Answer: The data support a positive difference, but the interval does not establish that the difference is at least 0.08., Explanation: Every plausible value is positive, so the interval supports $p_A > p_B$. However, values from 0.02 up to 0.08 are included, so the data do not establish that the difference meets the prespecified practical threshold.

APSTAT-MCQ11-003, Correct: B, Correct Answer: $1/3$, Explanation: Of the 6 possible assignments, exactly two are derangements: the card assignments cycle in either direction so that no person receives the matching card. Thus the probability is $2/6 = 1/3$.

APSTAT-MCQ11-004, Correct: A, Correct Answer: Form pairs of runners with similar baseline speeds, then randomly assign one runner in each pair to each routine and analyze within-pair recovery-time differences., Explanation: Matching controls the expected baseline-speed effect, and random assignment within each pair prevents routine from being confounded with speed. The resulting paired differences compare routines among similar runners.

APSTAT-MCQ11-005, Correct: D, Correct Answer: A paired leaf-area and transpiration-rate measurement from each sampled plant, Explanation: Regression requires two quantitative measurements associated with the same observational unit. Independently sorted lists or greenhouse-level averages destroy the plant-level pairing needed for prediction.

APSTAT-MCQ11-006, Correct: C, Correct Answer: Draw an SRS from the full ticket-holder database and contact the selected people after the festival., Explanation: The ticket database covers the stated population, including no-shows and early leavers. Exit and fan-page surveys systematically omit portions of that population or rely on voluntary response.

APSTAT-MCQ11-007, Correct: D, Correct Answer: 70 successes and (0.224, 0.336), Explanation: The success count is $250(0.28) = 70$. Applying the reported margin gives $0.28 \pm 0.056 = (0.224, 0.336)$.

APSTAT-MCQ11-008, Correct: C, Correct Answer: $P(X = 2) = 0.15$, Explanation: A CDF jump at 2 equals the probability mass at exactly 2. Therefore $P(X = 2) = F(2) - P(X < 2) = 0.45 - 0.30 = 0.15$.

APSTAT-MCQ11-009, Correct: B, Correct Answer: 1.5 units, Explanation: Squaring the total standard error gives $2.5^2 = 6.25$. The first contribution is $2.0^2 = 4$, so the second squared contribution is $6.25 - 4 = 2.25$ and $s_2/\sqrt{n_2} = \sqrt{2.25} = 1.5$ units.

APSTAT-MCQ11-010, Correct: D, Correct Answer: 0.47, Explanation: The cumulative proportion in the interval is $0.82 - 0.35 = 0.47$. Subtracting removes observations at or below 10 from those at or below 20.

APSTAT-MCQ11-011, Correct: A, Correct Answer: It is at most 64., Explanation: Under the old fitted line, removing the point subtracts $6^2 = 36$ from SSE, leaving 64 for the remaining observations. Refitting by least squares cannot produce a larger SSE than using that old line, so the new SSE is at most 64.

APSTAT-MCQ11-012, Correct: B, Correct Answer: p , the population proportion of all warranty claims filed within 30 days, Explanation: The goal is a population proportion, conventionally denoted p . The count x and sample proportion $\hat{p}$ are sample statistics, while μ denotes a population mean.

APSTAT-MCQ11-013, Correct: D, Correct Answer: Generate 25 integers from 1 to 365 with replacement, record whether any value repeats, and repeat for many trials., Explanation: Independent equally likely birthdays are represented by independent integers generated with replacement. A repeated value corresponds to at least two students sharing a birthday, and many trials approximate the probability.

APSTAT-MCQ11-014, Correct: A, Correct Answer: The mean is an arithmetic balance point but may poorly represent a typical observation when it lies in the gap between clusters., Explanation: A mean can fall between two clusters even when few or no observations occur near it. The two clusters are important features that a single center cannot describe, so the mean alone is not a representative typical value.

APSTAT-MCQ11-015, Correct: A, Correct Answer: Approximately 3.04, Explanation: The interval margin is $(5.8 - 1.2)/2 = 2.3$, so $SE = 2.3/2.00 = 1.15$. For the zero null, $t = 3.5/1.15 \approx 3.04$.

APSTAT-MCQ11-016, Correct: A, Correct Answer: The analysis can misrepresent the population proportion and its uncertainty unless sampling weights and the design are incorporated., Explanation: Deliberate disproportionate sampling changes inclusion probabilities. An unweighted statistic overrepresents rural households, and a simple-SRS standard error ignores the stratified unequal-probability design.

APSTAT-MCQ11-017, Correct: C, Correct Answer: Randomly selected transaction as the unit and seconds from joining the line to beginning checkout as the variable, Explanation: Each transaction generates one quantitative delay measurement, creating a distribution across transactions. Store totals, identification numbers, and opening status do not measure checkout-delay variation.

APSTAT-MCQ11-018, Correct: C, Correct Answer: $0.8^6 = 0.262144$, Explanation: All six independent checks must pass, so $P(X = 6) = C(6, 6)(0.8)^6(0.2)^0 = 0.8^6 = 0.262144$.

APSTAT-MCQ11-019, Correct: D, Correct Answer: $6(0.8^5)(0.2) + 0.8^6 = 0.65536$, Explanation: The event $X \geq 5$ consists of the disjoint outcomes $X = 5$ and $X = 6$. Their probabilities are $6(0.8^5)(0.2) = 0.393216$ and $0.8^6 = 0.262144$, totaling 0.65536.

APSTAT-MCQ11-020, Correct: B, Correct Answer: Roughly 655 of the 1,000 procedures will record five or six passing checks., Explanation: Multiplying 1,000 repetitions by 0.65536 gives about 655.36 procedures with $X \geq 5$, so roughly 655 is reasonable. This long-run expectation is neither an exact guarantee nor a claim about the fraction of individual checks that pass.

APSTAT-MCQ11-021, Correct: D, Correct Answer: Positive and negative residuals balance algebraically, but their counts and individual magnitudes need not match., Explanation: A zero residual sum constrains only the signed total. A few large residuals of one sign can balance many smaller residuals of the opposite sign, so equal counts or pairwise matching are not required.

APSTAT-MCQ11-022, Correct: A, Correct Answer: Randomly assign each eligible user to one banner for the study and record one prespecified purchase outcome per user., Explanation: User-level assignment and one outcome per user make the observational units distinct people. Treating repeated page views as independent would give frequent visitors disproportionate weight and understate dependence.

APSTAT-MCQ11-023, Correct: A, Correct Answer: 6, Explanation: Removing 1 and 40 leaves 4, 5, 6, and 9. Their mean is $(4 + 5 + 6 + 9)/4 = 24/4 = 6$.

APSTAT-MCQ11-024, Correct: D, Correct Answer: $\mu_A - \mu_B$, the difference in population mean classroom-average scores under curricula A and B, Explanation: Classrooms, not individual students, are the independently assigned experimental units. Defining the parameter through population classroom-average responses aligns the inferential unit with the randomization.

APSTAT-MCQ11-025, Correct: B, Correct Answer: 0.3174, Explanation: The upper-tail area beyond 1 is $1 - \Phi(1) = 0.1587$, while the conditioning region $Z > 0$ has area 0.5. Because $Z > 1$ is contained in $Z > 0$, the required conditional proportion is $0.1587/0.5 = 0.3174$.

APSTAT-MCQ11-026, Correct: B, Correct Answer: 99.5%, Explanation: The 99% margin is $2.576(0.030) = 0.07728$, slightly less than 0.080, so 0 remains outside. The 99.5% margin is $2.807(0.030) = 0.08421$, so the interval reaches below 0 and includes it.

APSTAT-MCQ11-027, Correct: A, Correct Answer: Use professional translation and independent back-translation, pilot the versions, and train interviewers to follow equivalent scripts., Explanation: Translation, independent checking, piloting, and standardized delivery help preserve meaning and measurement across languages. Improvised or conceptually different wording creates systematic measurement differences.

APSTAT-MCQ11-028, Correct: B, Correct Answer: 2.12, Explanation: The estimated standard error is $\sqrt{(4 + 4)} = \sqrt{8} \approx 2.828$ minutes. Centering the observed contrast at the null gives $t = (8 - 2)/2.828 \approx 2.12$.

APSTAT-MCQ11-029, Correct: A, Correct Answer: The data support a higher population mean for the new material, although individual strengths from the two materials may overlap., Explanation: Every value in the interval is positive, supporting $\mu_{\text{new}} > \mu_{\text{standard}}$. The interval concerns the difference between population averages; it neither orders every individual specimen nor determines whether the improvement meets every practical requirement.

APSTAT-MCQ11-030, Correct: C, Correct Answer: Range = 10 and IQR = 7, Explanation: Replacing the maximum 30 with 12 makes the new maximum 12 while the minimum remains 2, so the range is 10. The quartiles remain 5 and 12, so $\text{IQR} = 12 - 5 = 7$.

APSTAT-MCQ11-031, Correct: C, Correct Answer: $X = 1$ if the selected account is overdue and $X = 0$ otherwise., Explanation: For an indicator variable, $E(X) = 1 \cdot P(\text{overdue}) + 0 \cdot P(\text{not overdue}) = P(\text{overdue}) = p$. The other variables do not have expectation equal to the population proportion.

APSTAT-MCQ11-032, Correct: B, Correct Answer: The directional 5% rule rejects p_0 , while the 95% two-sided interval contains p_0 because it uses the larger cutoff., Explanation: The statistic 1.74 exceeds 1.645, so it passes the directional rejection boundary. It does not exceed 1.960, so the margin used by the two-sided 95% interval is large enough for that interval to reach p_0 .

APSTAT-MCQ11-033, Correct: D, Correct Answer: The two centers are equal, but the distribution can still be asymmetric or otherwise irregular., Explanation: Equality of mean and median alone does not determine the full distribution shape, spread, or symmetry. Many asymmetric datasets can have equal numerical values for these two centers.

APSTAT-MCQ11-034, Correct: B, Correct Answer: A paired t procedure can be reasonable because its condition concerns the distribution of within-pair differences., Explanation: A paired analysis reduces the data to one quantitative difference per pair. The relevant shape and outlier condition applies to those 15 differences, not separately to the before and after marginal distributions.

APSTAT-MCQ11-035, Correct: D, Correct Answer: $\hat{y} = 4 + 2x$, Explanation: Copying every point gives each original pair the same additional weight. Means, standard deviations, correlation, and therefore the least-squares coefficients remain unchanged.

APSTAT-MCQ11-036, Correct: A, Correct Answer: 0.60, Explanation: Exact duplication leaves the joint pattern and standardized cross-products unchanged, so the correlation remains 0.60. Copies do not strengthen the descriptive association.

APSTAT-MCQ11-037, Correct: C, Correct Answer: SSE becomes 192 because every squared residual is counted twice, but the copied rows provide no new independent measurements., Explanation: The fitted line and each residual value are unchanged, but every squared residual appears twice, so SSE doubles from 96 to 192. Exact copies add no independent observations.

APSTAT-MCQ11-038, Correct: C, Correct Answer: \$1.00, Explanation: Weighting each possible adjustment by its probability gives $E(Y) = 6(0.25) + 1(0.50) - 4(0.25) = 1.50 + 0.50 - 1.00 = \1.00 .

APSTAT-MCQ11-039, Correct: A, Correct Answer: 96%, Explanation: A two-sided p-value of 0.04 means the observed statistic places 0.02 in each tail beyond $\pm|z|$. The central area between those cutoffs is $1 - 0.04 = 0.96$, so the null is an endpoint of the corresponding 96% interval.

APSTAT-MCQ11-040, Correct: C, Correct Answer: The data provide nearly the same strength of evidence; a 0.05 convention changes the formal labels, not the evidence abruptly., Explanation: Tail areas 0.048 and 0.052 are numerically close and indicate nearly equal incompatibility with their respective null models. The conventional cutoff can alter a decision label without creating a discontinuity in evidential strength.

APSTAT-MCQ11-041, Correct: B, Correct Answer: Stratify by shift and randomly sample packages in proportion to each shift's share of total output., Explanation: Random sampling within every shift covers production conditions, and proportional allocation represents each shift according to its actual output. Convenience choices or one shift alone can bias the overall estimate.

APSTAT-MCQ11-042, Correct: B, Correct Answer: Randomly assign participants to indistinguishable diffusers containing the scent or an odor-matched inactive control, and blind participants and outcome scorers when feasible., Explanation: Random assignment creates comparable groups, the inactive control separates scent from treatment expectations, and blinding reduces expectancy and assessor effects.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ11

· APSTAT-MCQ11-001 to 005: Unit 1 tied value 7 at ordered positions 12–15 among 20 observations distinguishes 75% at most 7 from 55% below 7; Unit 3 workflow-proportion interval (0.02, 0.10) supports a positive effect without establishing the 0.08 practical threshold; Unit 2 a three-person derangement counts two no-match assignments among six; Unit 4 runners matched on baseline speed receive randomized warm-up routines; Unit 5 plant-level leaf area and transpiration require paired measurements per plant. Misconceptions include ignoring ties in percentile counts, equating statistical positivity with practical importance, counting only one cyclic derangement, confounding routine with baseline speed, and using unpaired marginal lists or greenhouse aggregates for regression.

· APSTAT-MCQ11-006 to 010: Unit 1 a festival ticket-holder SRS covers no-shows and early leavers; Unit 3 $\hat{p} = 0.28$ with $n = 250$ and margin 0.056 gives 70 successes and interval (0.224, 0.336); Unit 2 a CDF jump from 0.30 to 0.45 assigns point mass 0.15 at $X = 2$; Unit 4 total two-mean SE 2.5 and first component 2.0 imply second component 1.5; Unit 1 cumulative proportions 0.35 and 0.82 yield 0.47 in (10, 20]. Misconceptions include exit-only coverage, treating $\hat{p}$ as a count, confusing a CDF value with a jump, subtracting standard errors rather than squared contributions, and reading a cumulative endpoint without differencing.

· APSTAT-MCQ11-011 to 015: Unit 5 removing a residual-6 point from SSE 100 leaves old-line SSE 64 and refitting can only lower it; Unit 3 a 30-day warranty target defines population proportion p ; Unit 2 a 25-birthday collision simulation samples integers 1–365 independently with replacement; Unit 1 a two-cluster dotplot has mean 20 in a gap, showing why one center may not describe a typical observation; Unit 4 interval (1.2, 5.8), estimate 3.5, and $t^* = 2$ recover SE 1.15 and $t \approx 3.04$ against zero. Misconceptions include assuming refitted SSE must equal the old-line remainder, using sample statistics as parameters, sampling birthdays without replacement, requiring a mean to be observed or treating two clusters as unimodal, and dividing the estimate by the interval margin.

· APSTAT-MCQ11-016 to 020: Unit 3 fourfold rural oversampling requires weights and design-aware uncertainty; Unit 1 checkout delay is seconds per randomly selected transaction; Unit 2 a Binomial(6, 0.80) calibration set calculates all-six probability 0.262144, at-least-five probability 0.65536, and translates that event rate to roughly 655 among 1,000 repeated procedures. Misconceptions include

treating disproportionate samples as self-weighting, using store-level totals instead of transaction delays, omitting the binomial coefficient for five passes, excluding six from an at-least-five event, and treating a long-run expectation as an exact guarantee.

- APSTAT-MCQ11-021 to 025: Unit 5 zero residual sum means algebraic balance rather than equal positive and negative counts; Unit 3 banner assignment occurs once per user with one outcome to avoid repeat-view dependence; Unit 1 a defined trimmed mean removes 1 and 40 from six ordered values to give 6; Unit 4 classroom-level curriculum assignment targets the difference in population mean classroom-average scores; Unit 2 conditional standard-normal probability beyond $z = 1$ among observations above $z = 0$ divides 0.1587 by 0.5 to obtain 0.3174. Misconceptions include pairing residuals by sign, treating page views as independent people, retaining an extreme in the trimmed total, defining the parameter at the student rather than assignment-unit level, and using an unconditional upper tail after conditioning.

- APSTAT-MCQ11-026 to 030: Unit 3 estimate 0.080 and SE 0.030 first include zero at the listed 99.5% confidence level; Unit 1 multilingual survey versions use translation, back-translation, piloting, and standardized scripts; Unit 4 an observed curing-time contrast 8, null contrast 2, and squared-SE terms 4 and 4 give $t \approx 2.12$; Unit 4 tensile-strength interval (1.2, 4.8) supports a higher population mean without ordering every specimen; Unit 1 replacing maximum 30 by $Q3 = 12$ changes range to 10 while IQR stays 7. Misconceptions include overlooking the 99% margin just below 0.080, allowing improvised translation, failing to center a statistic at the hypothesized contrast, extending a mean claim to every specimen, and changing IQR because of an altered maximum.

- APSTAT-MCQ11-031 to 035: Unit 2 a 0–1 overdue-account indicator has expectation equal to the overdue proportion; Unit 3 $z = 1.74$ clears the directional cutoff 1.645 but not the two-sided-interval cutoff 1.960; Unit 1 equal mean and median do not force symmetry; Unit 4 paired t conditions apply to the within-pair differences rather than skewed marginal measurements; Unit 5 exact duplication of 12 regression rows leaves $\hat{y} = 4 + 2x$ unchanged. Misconceptions include using a count or waiting time instead of an indicator, treating distinct critical boundaries as contradictory, inferring shape from two centers, checking the wrong distributions for pairing, and doubling regression coefficients with row duplication.

- APSTAT-MCQ11-036 to 040: Unit 5 duplicated regression rows leave $r = 0.60$ but double SSE from 96 to 192 without adding independent information; Unit 2 catered-order adjustments +\$6, +\$1, and -\$4 yield expected value \$1.00; Unit 3 a two-sided p -value 0.04 places the null at the endpoint of a 96% interval; Unit 3 null-model tail areas 0.048 and 0.052 provide nearly identical evidence despite opposite threshold labels. Misconceptions include scaling correlation with row count, leaving SSE fixed when residuals are copied, averaging possible monetary outcomes without probability weights, matching $p = 0.04$ to 95% rather than central area 0.96, and treating 0.05 as a sharp scientific boundary.

- APSTAT-MCQ11-041 to 042: Unit 1 factory packages are proportionally stratified by unequal-output work shifts; Unit 1 a scent experiment uses randomized indistinguishable active and odor-matched inactive diffusers with participant and scorer blinding. Misconceptions include equal unweighted convenience samples across unequal shifts and confounding scent with expectation, time of day, or self-selection.

- Unit counts for APSTAT-MCQ11: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: Binomial(6, 0.80) calibration checks, APSTAT-MCQ11-018 to 020; exact duplication of a 12-pair regression dataset, APSTAT-MCQ11-035 to 037.